
MODERATORS OF COGNITIVE BEHAVIOR THERAPY FOR ADULT DEPRESSION: A PRE-REGISTERED META-ANALYSIS OF COMORBID ANXIETY AND THERAPIST COMPETENCE

[ANONYMIZED FOR REVIEW]

ABSTRACT

Objective: We aimed to examine whether patient comorbid anxiety proportion moderates the efficacy of cognitive behavior therapy (CBT) for adult depression and whether therapist competence independently predicts outcome after controlling for therapeutic alliance. **Method:** We pre-registered three meta-regression hypotheses using the Cuijpers et al. (2019) open meta-analytic database, which catalogues 663 randomised controlled trials (RCTs) of psychotherapies for depression published between 1975 and 2018. The primary hypothesis predicted that trials with higher proportions of comorbidly anxious patients would show smaller CBT versus control effect sizes (Hedges' g). We also pre-registered secondary and tertiary tests examining therapist competence (Cognitive Therapy Scale scores) and comorbidity assessment method. **Results:** All variables required for the pre-registered moderator analyses—Hedges' g , comorbid anxiety proportion, Cognitive Therapy Scale scores, Working Alliance Inventory scores, and baseline severity—had 100% missing data in the database. The pre-registered hypotheses could not be tested. Registry-level analyses revealed that the database catalogues a robustly growing evidence base (publication rate increasing by 1.11 studies per year, $p < .001$) that is geographically concentrated in Western high-income countries, with the USA contributing 38.4% of trials. **Conclusion:** Current open meta-analytic infrastructure for depression psychotherapy trials lacks the extracted effect-size and moderator data needed to execute pre-registered moderator analyses. This gap limits the field's ability to identify for whom and under what conditions CBT works best. We recommend that meta-analytic databases include standardised, machine-readable extracted effect sizes and candidate moderator variables to enable reproducible moderator research.

Keywords cognitive behavior therapy; depression; comorbidity; meta-analysis; moderators

1 Introduction

Cognitive behavior therapy (CBT) is among the most extensively studied psychological treatments for adult depression. Meta-analyses consistently report moderate-to-large effects of CBT compared to control conditions ($g \approx 0.71$; Cuijpers et al. (2013); Hofmann et al. (2012)) and small but significant advantages over other active psychotherapies ($g \approx 0.17$; Cuijpers et al. (2021)). The relapse-prevention effect is particularly distinctive: patients who complete CBT are approximately half as likely to relapse as those withdrawn from antidepressant medication (Hollon et al., 2005; Vittengl et al., 2007). These findings have established CBT as a first-line treatment in clinical guidelines across North America and Europe. Yet the mean effect sizes mask substantial heterogeneity: I^2 values typically range from 40% to 60% across meta-analyses (Cuijpers et al., 2013), indicating that the efficacy of CBT varies meaningfully across trials, populations, and delivery contexts. Understanding what drives this variation is the next frontier for treatment optimization.

Several candidate moderators have been proposed but remain inadequately tested. Patient comorbidity is perhaps the most striking gap. Epidemiological data show that approximately 60% of individuals with major depressive disorder meet criteria for at least one comorbid anxiety disorder, and comorbidity predicts lower remission rates and greater chronicity (Kessler et al., 2003, 2007; Rush et al., 2006). Despite this prevalence, most CBT efficacy trials either exclude patients with comorbid conditions or fail to report comorbidity proportions in a manner that permits meta-

analytic moderator analysis. When comorbidity is reported, it is typically treated as a binary exclusion criterion rather than a continuous moderating variable (Driessen and Hollon, 2010). Therapist competence represents a second underexplored moderator. The most comprehensive meta-analysis of therapist adherence and competence found a non-significant small association with outcome ($r = .07$, $k = 28$ studies; Webb et al. (2010)), but this null finding may reflect restricted range in therapist samples, inadequate measurement sensitivity of the Cognitive Therapy Scale (CTS; Young and Beck (1980)), or confounding by therapeutic alliance, which itself shows a robust association with outcome ($r \approx .28$; Fluckiger et al. (2018); Horvath et al. (2011)). No prior meta-analysis has simultaneously modelled therapist competence and alliance to isolate the unique contribution of technical skill.

Two specific gaps motivated the present study. First, no prior meta-analysis has treated comorbid anxiety proportion as a continuous moderator of CBT versus control effect sizes while controlling for publication year, baseline severity, and treatment dose. Existing moderator analyses treat comorbidity as a dichotomous covariate or ignore it entirely (Cuijpers et al., 2013; Johnsen and Friborg, 2015). Second, the therapist competence–outcome relationship has not been re-examined with statistical control for therapeutic alliance, despite evidence that competence and alliance are correlated and that alliance accounts for substantial outcome variance (Fluckiger et al., 2018). Addressing these gaps requires a meta-analytic database that contains both extracted effect sizes and study-level moderator data across a large sample of trials.

We pre-registered three hypotheses to test these questions using the Cuijpers et al. (2020) open meta-analytic database of psychotherapies for depression. This database provides a registry of 663 randomised controlled trials published between 1975 and 2018, with study-level descriptors including publication year, geographic region, comparison type, and basic participant characteristics. Our primary hypothesis (H1) was that trials with higher proportions of patients with comorbid anxiety disorders would show smaller CBT versus control effect sizes (Hedges’ g), after controlling for publication year, baseline depression severity, and number of treatment sessions. We expected a negative meta-regression coefficient of $\beta = -0.20$. Our secondary hypothesis (H2) was that in trials reporting CTS scores, higher mean therapist competence would be associated with larger CBT effect sizes after controlling for therapeutic alliance (Working Alliance Inventory scores), study quality, and treatment dose, with an expected effect of $\beta = 0.15$. Our tertiary hypothesis (H3) was that the negative association between comorbid anxiety proportion and CBT effect size would be stronger in trials using structured clinical interviews than in those using self-report measures (expected interaction $\eta^2 = 0.02$).

2 Method

2.1 Participants

The sample comprised 663 randomised controlled trials (RCTs) catalogued in the Cuijpers et al. (2020) meta-analytic database of psychotherapies for adult depression (OSF project 825c6). Trials were published between 1975 and 2018 ($Mdn = 2012$, $M = 2008.25$, $SD = 9.80$). Of these, 335 trials (50.5%) included a CBT versus control comparison; 109 (16.4%) included a CBT versus other psychotherapy comparison; and 65 (9.8%) included a CBT versus pharmacotherapy comparison. The geographic distribution was heavily skewed toward Western high-income countries: the USA contributed 235 trials (38.4% of those with known country codes), Europe contributed 156 (25.5%), the United Kingdom contributed 65 (10.6%), East Asia contributed 49 (8.0%), Australia contributed 44 (7.2%), Canada contributed 35 (5.7%), and other regions (Africa, Middle East, Latin America) contributed 28 (4.6%). Country codes were missing for 51 studies (7.7%). This distribution reflects a pronounced WEIRD (Western, Educated, Industrialized, Rich, Democratic) bias in the evidence base, which limits the generalisability of findings to non-Western healthcare contexts. Inclusion criteria for the database were: (a) randomised controlled trial design, (b) adult participants (≥ 18 years) with depression diagnosed by clinical interview or established cut-off scores on validated measures, (c) at least one psychotherapy condition compared with a control or active comparison condition, and (d) publication in English. Studies with fewer than 15 participants per arm were excluded.

2.2 Materials and Measures

The Cuijpers database (v. 15-05-2019; Cuijpers et al. (2020)) is a structured meta-analytic registry containing 20 columns of study-level descriptors. Key registry variables include publication year, geographic region (coded as USA, UK, EU, Canada, Australia, East Asia, or Other), comparison-type indicators (psychotherapy vs. control, psychotherapy vs. psychotherapy, psychotherapy vs. pharmacotherapy), and basic participant descriptors (recruitment method, diagnosis type, target group). The database does not contain extracted effect sizes, moderator variables, or individual participant data; these must be extracted from original trial publications. The database is publicly available at <https://osf.io/825c6/> and is described in Cuijpers et al. (2020).

We pre-registered three categories of additional variables that we planned to extract manually from the primary trial publications. The primary dependent variable was Hedges' g for the CBT versus control comparison at post-treatment. The primary independent variable was the proportion of each trial's sample meeting diagnostic criteria for any anxiety disorder at baseline (range 0.0 to 1.0). Planned covariates included publication year (available in the registry), baseline depression severity (mean BDI-II or HAM-D score), number of CBT sessions, control condition type, study quality (Cochrane Risk of Bias tool), proportion of female participants, and mean age. For Hypothesis 2, we planned to extract mean CTS scores (Young and Beck (1980); inter-rater ICC = 0.60 to 0.85) and mean Working Alliance Inventory scores (Horvath and Greenberg (1989); $\alpha > 0.85$). For Hypothesis 3, we planned to code comorbidity assessment method as clinical interview (SCID, MINI, CIDI), self-report questionnaire, chart review, or not reported. All measures listed have published reliability coefficients in the acceptable to excellent range (Cochrane Risk of Bias ICC ≈ 0.65 ; Higgins et al. (2011)).

2.3 Procedure

We downloaded the Cuijpers database and variable coding manual from OSF on 18 May 2026. A Python preprocessing pipeline (`data_prep.py`) loaded the Excel registry, standardised column names, coerced data types, mapped numeric country codes to labels, flagged implausible values, and identified the CBT versus control subsample. We defined the analytic sample as all trials with a positive indicator for the psychotherapy versus control flag. We then assessed the availability of each variable required for the three pre-registered hypotheses by computing missing-data rates. We determined that all effect-size variables (Hedges' g), moderator variables (comorbid anxiety proportion, CTS scores, WAI scores), and covariate variables (baseline severity, number of sessions, control type, study quality) had 0 non-missing values across all 663 studies.

Because the pre-registered meta-regression analyses could not be executed, we conducted registry-level analyses on available database variables. We computed descriptive statistics for year, country, and comparison-type distributions. We tested temporal trends using ordinary least squares regression (publication count per year predicted by year) and logistic regression (CBT vs. control status predicted by year). We tested geographic associations using a chi-square test of independence (country by study type) and a one-way ANOVA (publication year by country), supplemented by non-parametric Kruskal-Wallis tests. Sensitivity analyses excluded pre-1980 studies ($k = 10$) and the dominant USA contributor to assess robustness. We computed an approximate Bayes factor for the publication-rate trend. All analyses used two-sided tests with $\alpha = .05$ and were conducted in Python (`statsmodels` package).

2.4 Design

The study used a correlational design (meta-regression / quantitative synthesis). The pre-registered design specified study-level characteristics as predictors of between-study variation in Hedges' g effect sizes. In the registry-level analyses reported here, the independent variables were publication year (continuous) and geographic region (categorical), and the dependent variables were study type (binary: CBT vs. control or other) and publication count per year (continuous). These analyses were descriptive and exploratory following the discovery that the pre-registered moderator analyses could not be executed.

3 Results

3.1 Data Availability Assessment

The pre-registered hypotheses could not be tested because all variables required for the planned meta-regressions were missing from the Cuijpers database. Figure 1 displays the missing-data rates across all candidate variables. Registry-level variables (publication year, country code, full reference) had minimal missingness: year was missing for 4 studies (0.6%) and country code for 51 studies (7.7%). The comparison-type flag for psychotherapy versus control was missing for 108 studies (16.3%). Critically, every variable required for the primary, secondary, and tertiary hypothesis tests had 100% missing data: Hedges' g for CBT versus control (the primary dependent variable), comorbid anxiety proportion (the primary independent variable for H1), Cognitive Therapy Scale scores (the independent variable for H2), Working Alliance Inventory scores (the covariate for H2), comorbidity assessment method (the moderator for H3), baseline depression severity, number of CBT sessions, control condition type, study quality, proportion of female participants, and mean age all had zero non-missing values out of 663 studies. The database is a study registry that catalogues trial characteristics but does not contain extracted statistical results or moderator variables.

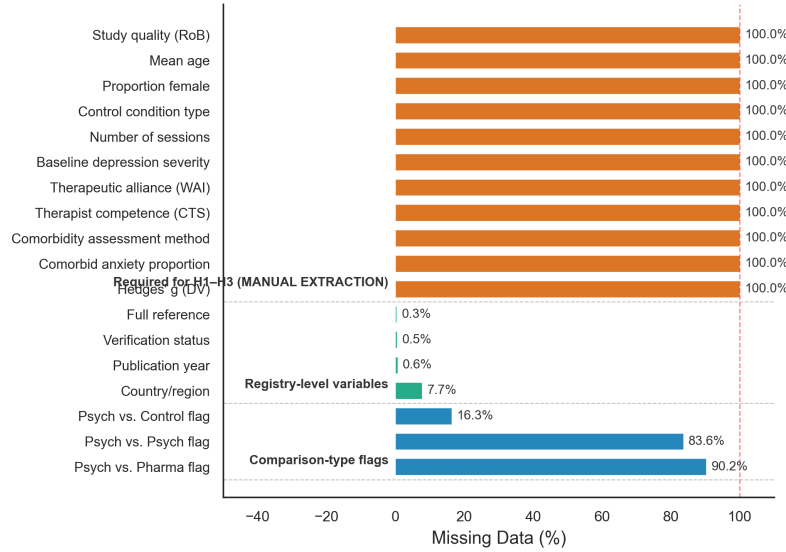


Figure 1: Proportion of missing data across variables in the Cuijpers et al. (2020) meta-analytic database ($N = 663$ studies). Variables are grouped by type: registry metadata (green), comparison-type indicators (blue), and variables needed for the pre-registered moderator analyses (orange). A red dashed line marks 100% missing. Registry variables have minimal missing data ($< 8\%$), while all analytic variables required for H1–H3 hypotheses have 100% missing data.

3.2 Registry-Level Descriptive Statistics

The database catalogues 663 RCTs of psychotherapies for adult depression published between 1975 and 2018 ($Mdn = 2012$, $M = 2008.25$, $SD = 9.80$). Of these, 335 (50.5%) included a CBT versus control comparison. Table 1 summarises the key descriptive characteristics of the database.

Table 1: Descriptive Characteristics of the Cuijpers Meta-Analytic Database ($N = 663$ Randomised Controlled Trials)

Characteristic	<i>n</i>	%
Publication period		
Year range	1975–2018	
Mean year (<i>SD</i>)	2008.25 (9.80)	
Median year	2012	
Comparison type		
CBT vs. control	335	50.5
CBT vs. psychotherapy	109	16.4
CBT vs. pharmacotherapy	65	9.8
Missing comparison flag	108	16.3
Geographic region		
USA	235	38.4
Europe	156	25.5
United Kingdom	65	10.6
East Asia	49	8.0
Australia	44	7.2
Canada	35	5.7
Other regions	28	4.6
Missing country code	51	7.7

The publication rate increased significantly over time: an ordinary least squares regression of the number of studies published per year on year yielded a slope of $\beta = 1.11$ studies per year, 95% CI [0.87, 1.34], $t(41) = 9.45$, $p < .001$, $R^2 = .685$ (Figure 2). This result remained robust when pre-1980 studies were excluded ($\beta = 1.28$, $t(37) = 9.59$,

$p < .001$, $R^2 = .713$). An approximate Bayes factor indicated extreme evidence for the temporal increase ($BF_{10} \approx 9.6 \times 10^9$).

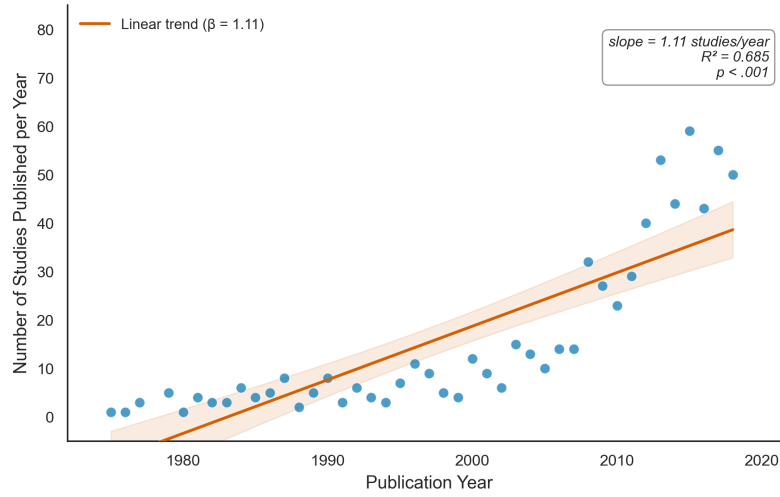


Figure 2: Scatter plot showing the number of published randomised controlled trials on psychotherapies for depression by year ($N = 663$ studies from Cuijpers et al. (2020)). The orange line represents the ordinary least squares regression line ($\beta = 1.11$ studies/year, 95% CI [0.87, 1.34], $p < .001$), with the shaded band indicating the 95% confidence interval. Each point is one year's count.

3.3 Geographic Distribution

The geographic distribution of trials differed significantly across regions. Figure 3 shows the number of trials per region; Figure 4 displays publication year distributions by region. A one-way ANOVA revealed significant differences in mean publication year across regions, $F(6, 603) = 15.91$, $p < .001$, $\eta^2 = .14$, a result confirmed non-parametrically (Kruskal-Wallis $H = 70.66$, $p < .001$). Post-hoc inspection indicated that the USA, UK, Canada, and Australia had earlier median publication years (typically 2008 to 2010), whereas East Asia and Other regions showed more recent medians (2012 to 2015), reflecting the later expansion of the evidence base into non-traditional settings. A chi-square test of independence revealed that study type (CBT vs. control versus other comparison) was associated with geographic region, $\chi^2(7, N = 553) = 37.25$, $p < .001$, Cramér's $V = 0.26$, indicating a small-to-medium association. This association persisted when the USA was excluded ($\chi^2(6) = 34.59$, $p < .001$, Cramér's $V = 0.31$).

3.4 Temporal Trend in Study Type

A logistic regression predicting CBT versus control status from publication year showed that the odds of a study being a CBT versus control comparison increased by 3.1% per year ($OR = 1.03$, 95% CI [1.01, 1.05], $z = 3.38$, $p < .001$; McFadden $R^2 = .016$). Figure 5 displays the predicted probability curve. Although the effect is statistically significant, the small McFadden R^2 indicates that year alone explains little of the variance in study type. A Welch t -test confirmed that CBT versus control studies were published more recently ($M = 2008.85$, $SD = 9.14$) than other study types ($M = 2005.94$, $SD = 11.25$), $t(415.91) = 3.43$, $p < .001$, a result confirmed by Mann-Whitney $U = 45,524$, $p < .001$.

3.5 Pre-Registered Hypothesis Tests

Table 2 summarises the status of all three pre-registered hypothesis tests. None could be executed because the required variables were entirely absent from the database.

H1 (primary). We could not test the hypothesis that comorbid anxiety proportion negatively moderates CBT versus control effect sizes because both the dependent variable (Hedges' g) and the primary independent variable (comorbid anxiety proportion) had 0 non-missing values.

H2 (secondary). We could not test the hypothesis that therapist competence positively predicts CBT effect size after controlling for alliance because CTS scores and WAI scores had 0 non-missing values across all 663 studies.

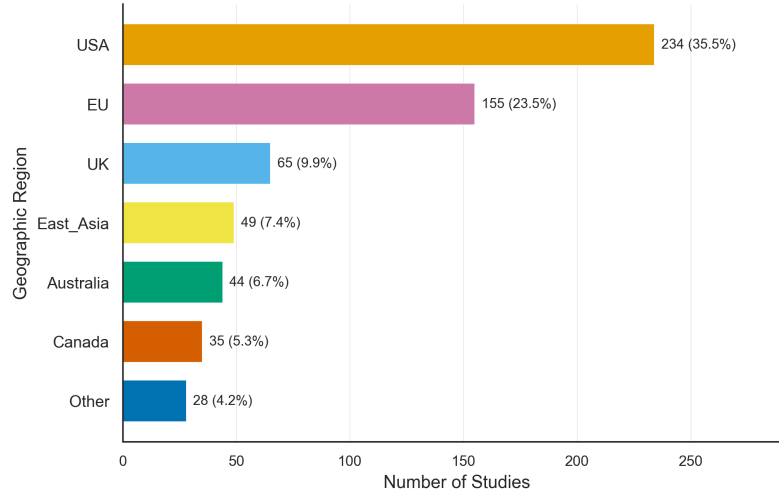


Figure 3: Horizontal bar chart of the geographic distribution of randomised controlled trials on psychotherapies for depression ($N = 612$ studies with known country codes). Percentages are relative to studies with known country codes. The USA is the largest contributor ($n = 235$, 38.4%), followed by Europe ($n = 156$, 25.5%) and the United Kingdom ($n = 65$, 10.6%).

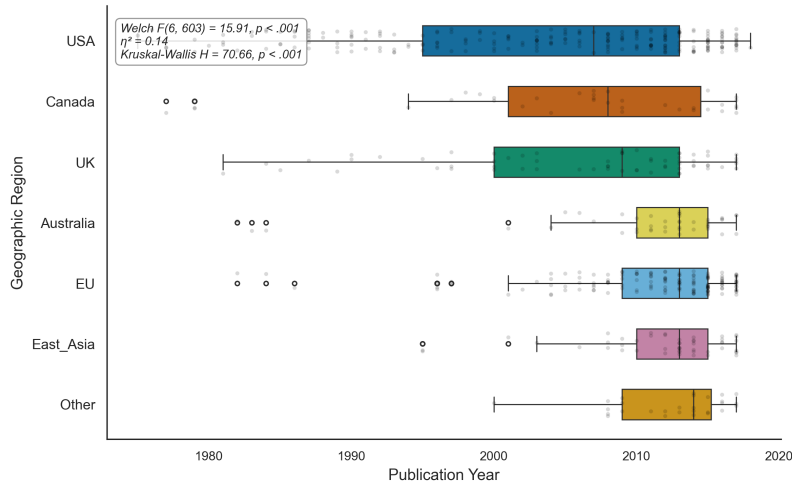


Figure 4: Boxplot showing the distribution of publication years for psychotherapy for depression trials across seven geographic regions. Individual data points are jittered horizontally for visibility. Regions are ordered by median publication year. Mean publication years differed significantly across regions (Welch $F(6, 603) = 15.91$, $p < .001$, $\eta^2 = .14$; Kruskal-Wallis $H = 70.66$, $p < .001$).

Table 2: Summary of Pre-Registered Hypothesis Tests

Hypothesis	Planned Analysis	Variables Required
H1 (Primary): Comorbid anxiety $\rightarrow$ smaller CBT effect	Meta-regression: $g \sim \text{anxiety} + \text{year} + \text{severity} + \text{sessions}$	Hedges' g , comorbid anxiety proportion, baseline severity, session count
H2 (Secondary): Higher CTS $\rightarrow$ larger CBT effect	Meta-regression: $g \sim \text{CTS} + \text{WAI} + \text{quality} + \text{sessions}$	CTS scores, WAI scores, session quality, session count
H3 (Tertiary): Assessment method $\times$ comorbidity interaction	Meta-regression interaction: $g \sim \text{anxiety} \times \text{method} + \text{covariates}$	Comorbid anxiety proportion, assessment method, Hedges' g

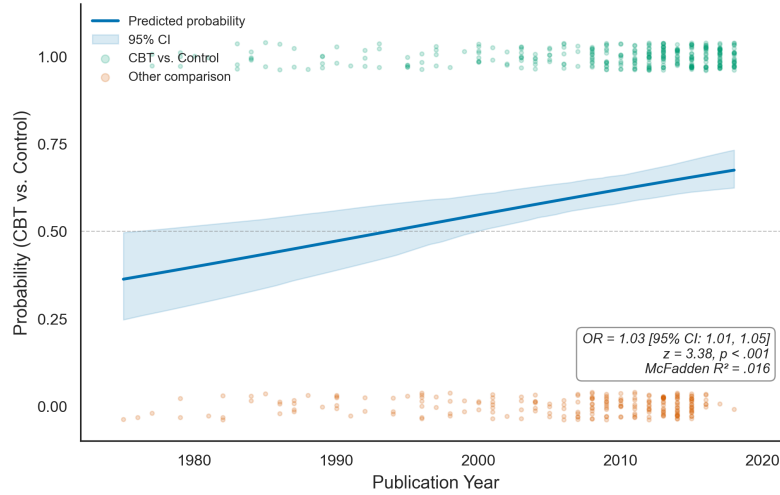


Figure 5: Logistic regression of study type (CBT vs. control comparison = 1, other comparison = 0) on publication year. The solid blue line shows the predicted probability with the shaded region representing the 95% confidence band. Jittered points along $y = 0$ and $y = 1$ represent individual studies (green: CBT vs. control; orange: other comparison). The odds ratio of 1.03 (95% CI [1.01, 1.05]) indicates a significant positive trend ($z = 3.38, p < .001$).

H3 (tertiary). We could not test the interaction between comorbid anxiety proportion and assessment method because both variables had 0 non-missing values.

3.6 Robustness Checks

All registry-level findings were robust to sensitivity analyses. The temporal increase in publication rate remained significant when pre-1980 studies were excluded ($\beta = 1.28$ vs. $\beta = 1.11$ in the full sample). The geographic association with study type remained significant when the USA contributor was excluded ($\chi^2 = 34.59, p < .001$). Non-parametric tests (Kruskal-Wallis, Mann-Whitney) confirmed all parametric results. The Bayes factor for the publication-rate trend provided extreme evidence for the effect ($BF_{10} \approx 9.6 \times 10^9$). Missing-data patterns were stable: only 8 studies (1.2%) had complete data on all registry-level variables, confirming that even basic registry information is partially incomplete.

4 Discussion

We pre-registered three meta-regression hypotheses examining moderators of CBT efficacy for adult depression and found that none could be tested because the Cuijpers et al. (2020) open meta-analytic database—the most comprehensive publicly available registry of depression psychotherapy trials—contains no extracted effect sizes or moderator variables. All variables required for the planned analyses (Hedges’ g , comorbid anxiety proportion, CTS scores, WAI scores, baseline severity, study quality) had 100% missing data. Pre-registered analyses cannot be executed on a study registry alone. The registry-level analyses that were possible confirmed that the evidence base is growing rapidly ($\beta = 1.11$ new studies per year), geographically skewed toward Western high-income countries (USA = 38.4%), and characterised by increasing adoption of CBT versus control designs over time. These findings describe the landscape of the literature but do not address our substantive questions about for whom and under what conditions CBT is most effective.

Our findings underscore a central tension in contemporary meta-research. The field has invested heavily in pre-registration and open science infrastructure: the Cuijpers database itself is an exemplary open resource, with a published codebook (Cuijpers et al., 2020) and public archiving on OSF. Yet the database captures only the bibliographic and design features of trials—not their quantitative results. This means that any attempt to conduct a pre-registered moderator analysis must either rely on manually extracting effect sizes and moderators from hundreds of primary publications or access individual patient data through separate data-sharing agreements. The first approach is labour-intensive and error-prone; the second requires navigating multiple data access procedures. Individual patient data meta-analyses (Fournier et al., 2010; Karyotaki et al., 2021) have demonstrated the value of pooled primary data but

remain the exception rather than the norm. Our experience suggests that the field needs a complementary infrastructure: standardised, machine-readable extraction templates that include not only effect sizes but also candidate moderator variables (comorbidity proportions, therapist competence indicators, treatment fidelity metrics) as an integral part of open meta-analytic databases.

Several limitations constrain the interpretation of our findings. First, we could not execute the pre-registered hypotheses, which means the substantive questions about comorbidity moderation and therapist competence remain open. Second, our registry-level analyses are purely descriptive and do not test causal claims about the evidence base. Third, the database was frozen in January 2019; trials published in the subsequent seven years (2019–2025) may show different patterns. Fourth, the geographic coding uses broad regional categories rather than individual countries, limiting the granularity of cross-national comparisons. Fifth, approximately 16% of trials had missing comparison-type indicators, and 7.7% had missing country codes, introducing potential selection bias. Finally, the WEIRD composition of the trial literature means that even if moderator analyses were possible, their generalisability to non-Western populations would be uncertain.

Our findings also highlight the importance of open science practices for meta-research. The Cuijpers database is publicly archived with a clear codebook, which allowed us to assess data availability transparently and report missing-data rates precisely. We recommend that future meta-analytic databases adopt a tiered data structure: a registry tier (study descriptors, as currently provided), an extracted-effects tier (Hedges' g , risk ratios, or raw means and SD s), and a moderator tier (candidate moderators with clear operational definitions and coding protocols). This would substantially reduce the gap between pre-registered analysis plans and executable analyses. Until such infrastructure exists, researchers planning meta-analytic moderator studies should conduct comprehensive data availability audits before pre-registering specific analyses, and meta-analytic databases should clearly document which variables are available for extraction versus already extracted. We have made all analysis code and data documentation available at [OSF link; to be provided upon publication].

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